

Santiago Hernández Senosiain

✉ cv@santihdzs.com |  santihdzs |  santihdzs |  portfolio

EDUCATION

Instituto Tecnológico y de Estudios Superiores de Monterrey – BSc Computer Science Aug. 2024-Present

- Relevant Coursework: Algorithms, Database Systems, Object-Oriented Programming.

The Edron Academy – International Baccalaureate Diploma

July 2024

- Higher Level Subjects: Computer Science, Mathematics, Economics.

PROFESSIONAL EXPERIENCE

OBI – Web Developer

May 2026 – Jul. 2026

- Built and deployed a B2B product catalog site (obimx.com) for a wholesale client.
- Implemented client-side catalog search and filtering, plus a quote-request flow via EmailJS, over client-managed data sources.

ITESM – Undergraduate Researcher

Oct. 2025 – Present

- Supervised by Dr. Gerardo Ibarra Vázquez
- Conducting robustness evaluation of DCNNs under adversarial perturbations.
- Built the experimental infrastructure for the project from scratch, including reproducible pipelines for running and tracking attack and evaluation experiments.

ITESM – Teaching Assistant

Aug. 2025 – Dec. 2025

- Assisted in the Programming of Data Structures and Fundamental Algorithms course (TC1031).
- Designed programming assignments and solution sets for ~90 students.
- Created presentations to support lectures on core data structures and algorithms.

EXTRACURRICULAR

ITESM Competitive Programming Club – Member

Aug. 2025 – Aug. 2026

- Attended regular problem-solving sessions focused on algorithms, data structures, and competitive programming techniques.
- Collaborated with a 3-member team to prepare for national and international competitions, including the International Collegiate Programming Contest (ICPC).

RESEARCH PROJECTS

RSA Encryption | *Number Theory, Algebraic Structures, Cybersecurity*

Mar. 2024

- Conducted in-depth research on the security of RSA encryption, identifying potential vulnerabilities in its implementation.
- Utilized group theory to analyze the algebraic structure of the cryptosystem, devising a theoretical approach to derive private keys from public keys.
- Authored a paper on the findings, which received the highest possible grade (7) for IB coursework.

National Hockey League Modeling | *Mathematical Modeling, Large-scale Data Analysis*

Oct. 2023

- Developed a mathematical model to predict winning teams in National Hockey League (NHL) matchups, based on the Bradley-Terry model.
- Enhanced predictions by implementing exponentially weighted moving averages (EWMA) to account for recent performance trends.

SKILLS, CONTRIBUTIONS & INTERESTS

Languages: English and Spanish (fluent in both)

Software: Python (PyTorch, TensorFlow), C++, SQL, JS | Git, Bash, Docker, AWS | Unix Environments

Open Source: Contributed to Flax (Google), the Python programming language, PyTorch, and Rust

Interests: Machine Learning, Cybersecurity, Data Science, Number Theory